Computational Neuroscience Study

Information Theory

Minsu Zhang

May 19, 2021

Outline

Book	Problems
1. Entropy and mutual information 2. Information and entropy maximization	1. Prob 1 2. Prob 8 3. Prob 9

Entropy and Mutual Information

Mutual Information

- Mutual information

$$I_{\mathrm{m}} = \sum_{s,r} P[s]P[r|s]\log_2\left(\frac{P[r|s]}{P[r]}\right). \qquad (4.9)$$

- Kullback-Leibler divergence
  - Measure of how one probability distribution is different from a second

$$D_{\mathrm{KL}}(P, Q) = \sum_{r} P[r]\log_2\left(\frac{P[r]}{Q[r]}\right). \qquad (4.15)$$

1. Mutual information is the KL divergence between the distributions P[r, s] and P[r]P[s].
2. If the stimulus and the response were independent of one another, P[r, s] would be equal to P[r]P[s].

Continuous variables

- Differential entropy
- The probability of the firing rate lying in the range between r and r+Δr, for small r, is expressed in terms of a probability density as p[r]Δr.

$$H = -\sum_r P[r]\log_2 P[r]\,. \qquad (4.3)$$

$$\begin{aligned} H &= -\sum p[r]\Delta r \log_2(p[r]\Delta r) \\ &= -\sum p[r]\Delta r \log_2 p[r] - \log_2 \Delta r\,. \end{aligned} \qquad (4.16)$$

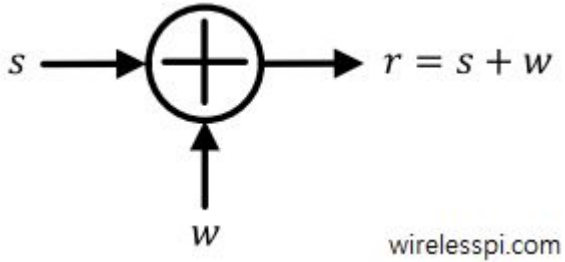
$$\lim_{\Delta r\to 0}\{H + \log_2 \Delta r\} = \boxed{-\int dr\, p[r]\log_2 p[r]}. \qquad (4.17)$$

- Noise entropy

$$\lim_{\Delta r\to 0}\{H_{\text{noise}} + \log_2 \Delta r\} = -\int ds \int dr\, p[s]p[r|s]\log_2 p[r|s]\,. \qquad (4.18)$$

Continuous variables

- Additive noise channel



- Mutual information

$$I_{\mathrm{m}} = \int ds \int dr\, p[s]p[r|s] \log_2 \left(\frac{p[r|s]}{p[r]} \right) . \qquad (4.19)$$

Information and Entropy Maximization

Single Neuron Entropy Maximization

- Computational implications of an observed response selectivity.

➔ Whether neural responses to natural stimuli are optimized to convey as much information as possible.

- Entropy maximization for a single neuron

$$-\int_0^{r_{\max}} dr\, p[r] \log_2 p[r]\,, \tag{4.20}$$

$$\int_0^{r_{\max}} dr\, p[r] = 1\,. \tag{4.21}$$

$$p[r] = \frac{1}{r_{\max}}\,, \tag{4.22}$$

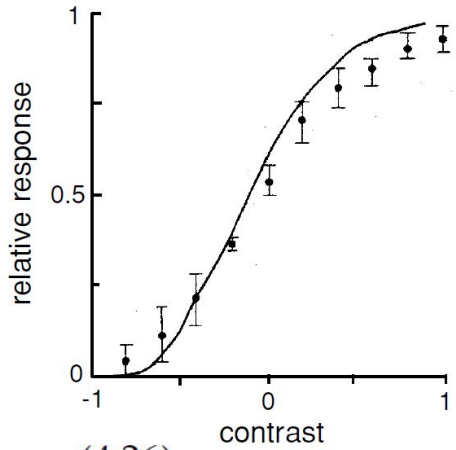
$$H = \log_2 r_{\max} - \log_2 \Delta r = \log_2 \left(\frac{r_{\max}}{\Delta r}\right)\,. \tag{4.23}$$

Neuronal Population Entropy Maximization

- Monotonically increasing response

$$\frac{df}{ds} = r_{\max} p[s] \,, \qquad (4.24)$$

$$f(s) = r_{\max} \int_{s_{\min}}^{s} ds' \, p[s'] \,, \qquad (4$$



rast response of the fly LMC (data points) compared to the inte-
bility distribution (solid curve). The relative re-
embrane potential fluctuation produced by the
h a given level of contrast divided by the max-
ed relative to the background level of illumina-
31.)

- Populations of neurons

$$H = -\int d\mathbf{r} \, p[\mathbf{r}] \log_2 p[\mathbf{r}] - N \log_2 \Delta r \,. \qquad (4.26)$$

$$H_a = -\int dr_a \, p[r_a] \log_2 p[r_a] - \log_2 \Delta r = -\int d\mathbf{r} \, p[\mathbf{r}] \log_2 p[r_a] - \log_2 \Delta r \,. \qquad (4.27)$$

$$\sum_a H_a - H = \int d\mathbf{r} \, p[\mathbf{r}] \log_2 \left(\frac{p[\mathbf{r}]}{\prod_a p_a[r_a]} \right) \geq 0 \,. \qquad (4.29)$$

$$H \leq \sum_a H_a \,. \qquad (4.28)$$

Achieving Entropy Maximization

- Factorial code

$$p[\mathbf{r}] = \prod_a p[r_a] \,, \tag{4.30}$$

- To achieve maximum population-response entropy:
  1. Factorization - the individual neurons must respond independently
  2. Probability equalization - they must all have response probabilities that are optimal for whatever constraints are imposed.

$$Q_{ab} = \int d\mathbf{r}\, p[\mathbf{r}](r_a - \langle r \rangle)(r_b - \langle r \rangle) = \sigma_r^2 \delta_{ab} \,. \tag{4.31}$$

Problem 4-8

Prob. 4-8

8. Suppose that a stimulus at one point on the retina, and at a given time, $s_r = s_s + \eta$, is the sum of a true stimulus s_s and a noise term η, as in exercise 6. Model the retinal processing at this particular location as producing a signal at the thalamus

$$s_l = D_s s_r + \eta_l ,$$

where D_s is a parameter called the transfer constant, and η_l represents an additional, independent source of noise that can be modeled as being drawn from a Gaussian distribution with mean 0 and variance $Q_{\eta_l\eta_l}$. Calculate the mutual information I_l between s_l and s_s as a function of D_s. The power of the signal produced by the retina is defined as $P_r = \langle (D_s s_r)^2 \rangle$. By maximizing

$$I_l - kP_r$$

as a function of D_s, find the transfer constant that maximizes the mutual information for a given value of k (with $k > 0$), a parameter that controls the trade-off between information and power. What happens when Q_{ss}, describing the visual signal, gets much smaller than $Q_{\eta\eta}$? (Based on a problem from Dawei Dong.)

@retina @thalamus

Prob 4-8 $s_r = s_s + \eta$ ① $s_l = D_s s_r + \eta_l$ ②

$I_l \sim f(D_s)$

Power: $P_r = \langle (D_s s_r)^2 \rangle$

$\arg\max_{D_s} (I_l - kP_r)$ $Q_{ss} < Q_{\eta\eta}$

$$I_l = \int ds_s \int ds_l \, p(s_s) p(s_l | s_s) \log_2 \left(\frac{p(s_l | s_r)}{p(s_l)} \right)$$

(s_l, s_r)

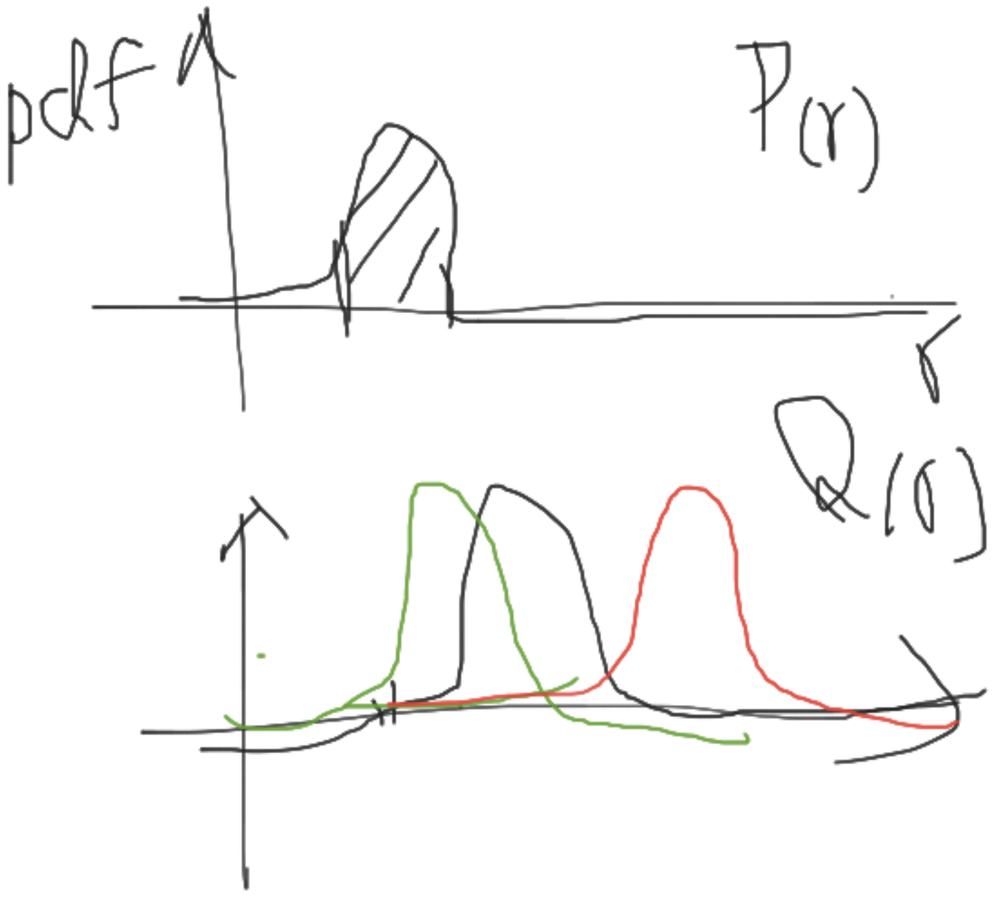
② ← ①: $s_l = D_s s_s + D_s \eta + \eta_l$

References

1. Dayan, P., & Abbott, L. F. (2001). Theoretical neuroscience: computational and mathematical modeling of neural systems. Computational Neuroscience Series.

$$I_{\mathrm{m}} = \sum_{s,r} P[s]P[r|s]\log_2\left(\frac{P[r|s]}{P[r]}\right). \tag{4.9}$$

$$\boxed{D_{\mathrm{KL}}(P, Q) = \sum_{r} P[r]\log_2\left(\frac{P[r]}{Q[r]}\right).} \tag{4.15}$$

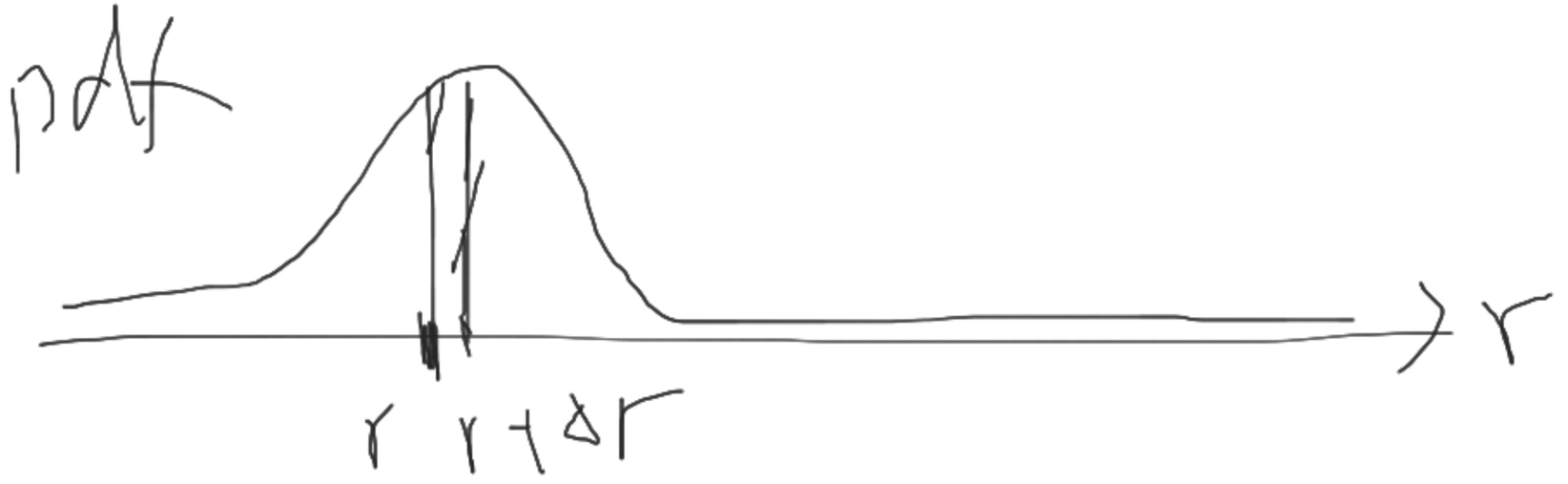


Mutual information is the KL divergence between the distributions P[rs] and P[r]P[s].

If the stimulus and the response were independent of one another, P[rs] would be equal to P[r]P[s].

Entropy of continuous variables

The probability of the firing rate lying in the range between r and r+Δr, for small r, is expressed in terms of a probability density as p[r]Δr.



$$H = -\sum_r P[r] \log_2 P[r] \,. \qquad (4.3)$$

$$H = -\sum_r P[r]\log_2 P[r]\,. \tag{4.3}$$

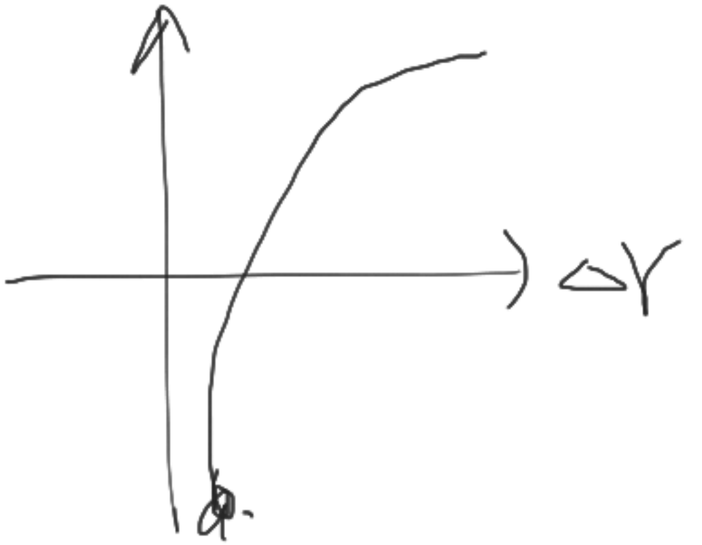
$$H = -\sum p[r]\Delta r \log_2 (p[r]\Delta r)$$

$$= -\sum p[r]\Delta r \log_2 p[r] - \log_2 \Delta r\,. \tag{4.16}$$

Differential entropy

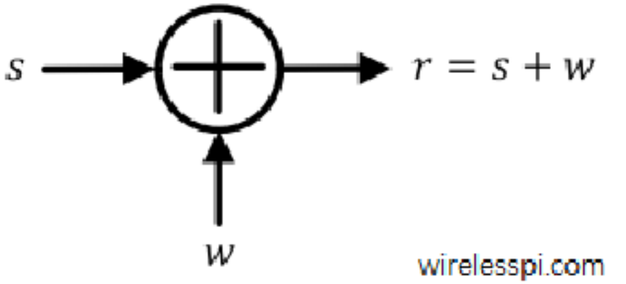
$$\lim_{\Delta r \to 0} \{H + \log_2 \Delta r\} = -\int dr\, p[r]\log_2 p[r]\,. \tag{4.17}$$

$$\lim_{\Delta r \to 0} \{H_{\text{noise}} + \log_2 \Delta r\} = -\int ds \int dr\, p[s]p[r|s]\log_2 p[r|s]\,. \tag{4.18}$$



Mutual information

$$I_{\text{m}} = \int ds \int dr\, p[s]p[r|s]\log_2 \left(\frac{p[r|s]}{p[r]}\right)\,. \tag{4.19}$$



Information and Entropy Maximization

Whether neural responses to natural stimuli are optimized to convey as much information as possible.

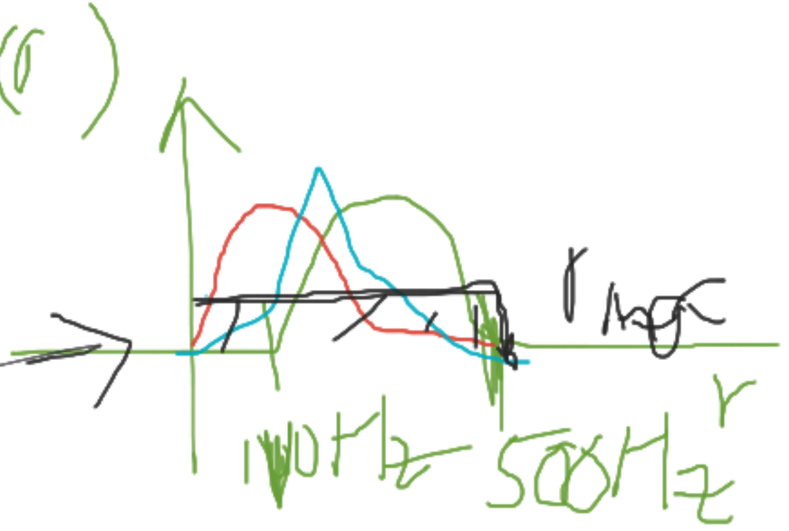
1) Entropy maximization for a single neuron

$$-\int_0^{r_{\max}} dr\, p[r] \log_2 p[r]\,, \tag{4.20}$$

$$\int_0^{r_{\max}} dr\, p[r] = 1\,. \tag{4.21}$$

$$p[r] = \frac{1}{r_{\max}}\,, \tag{4.22}$$

$$H = \log_2 r_{\max} - \log_2 \Delta r = \log_2\left(\frac{r_{\max}}{\Delta r}\right)\,. \tag{4.23}$$



$$H = \log_2 r_{\max} - \log_2 \Delta r = \log_2 \left(\frac{r_{\max}}{\Delta r} \right) . \tag{4.23}$$

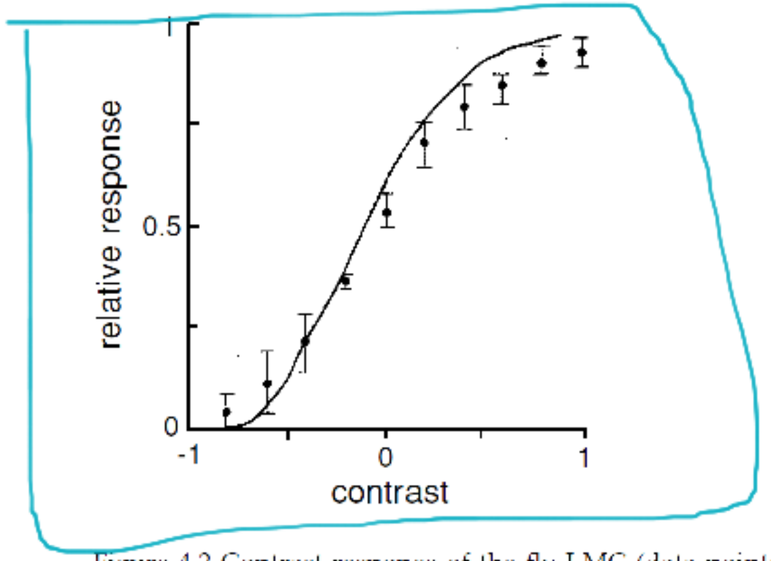


Figure 4.2 Contrast response of the fly LMC (data points) compared to the integral of the natural contrast probability distribution (solid curve). The relative response is the amplitude of the membrane potential fluctuation produced by the onset of a light or dark image with a given level of contrast divided by the maximum response. Contrast is defined relative to the background level of illumination. (Adapted from Laughlin, 1981.)

Monotonically increasing response

$$\frac{df}{ds} = r_{\max} p[s] , \tag{4.24}$$

$$f(s) = r_{\max} \int_{s_{\min}}^{s} ds' \, p[s'] , \tag{4.25}$$

s

$r = f(s)$

Populations of neurons

$$H = -\int d\mathbf{r} \, p[\mathbf{r}] \log_2 p[\mathbf{r}] - N \log_2 \Delta r . \tag{4.26}$$

$$\int_0^{r_{\max}} dr \, p[r] = 1 . \tag{4.21}$$

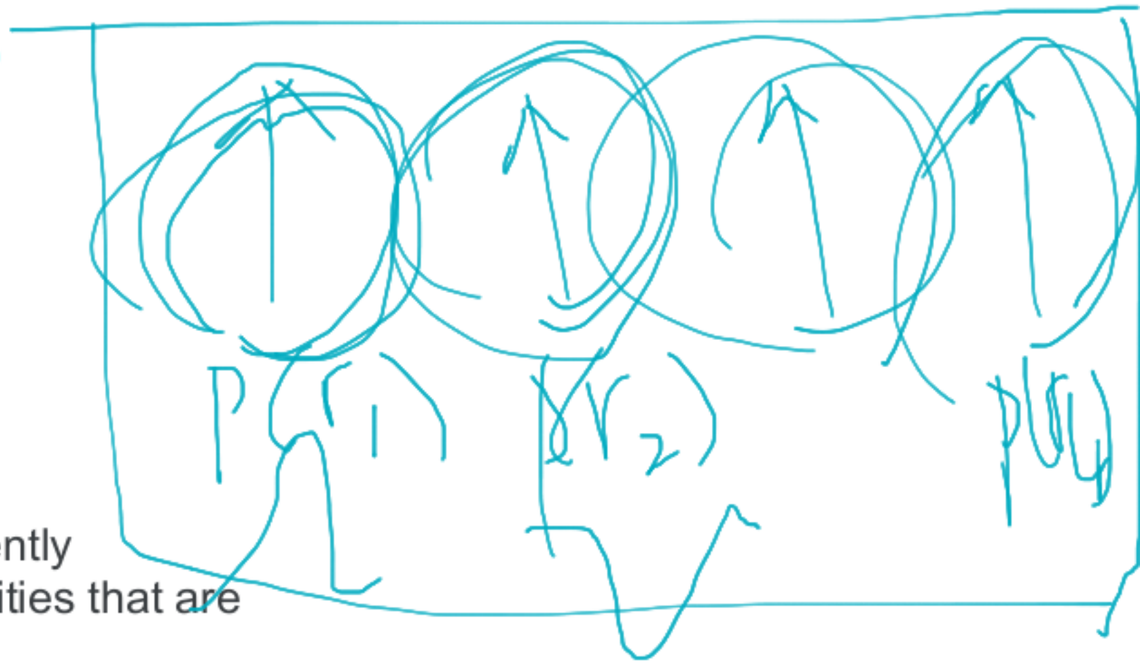
$$p[r] = \frac{1}{r_{\max}} ,$$

$$H = \log_2 r_{\max} - \log_2 \Delta r = \log_2 \left(\frac{r_{\max}}{\Delta r} \right) .$$

Factorial code

$$p[\mathbf{r}] = \prod_a p[r_a] \,, \tag{4.30}$$

$$p(a,b) = p(a)\cdot p(b)$$



To achieve maximum population-response entropy:
1) Factorization - the individual neurons must respond independently
2) Probability equalization - they must all have response probabilities that are optimal for whatever constraints are imposed.

$$Q_{ab} = \int d\mathbf{r}\, p[\mathbf{r}](r_a - \langle r\rangle)(r_b - \langle r\rangle) = \sigma_r^2 \delta_{ab} \,. \tag{4.31}$$